

Siddhant Choudhary

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EDUCATION

University of Wisconsin–Madison

Madison, WI

B.S. in Computer Science and Mathematics | GPA: 3.6/4.0

May 2028

- **Coursework:** Data Structures & Algorithms, Machine Organization and Programming, Introduction to Big Data Systems (CS 544), Linear Algebra, Probability & Statistical Inference
- **Leadership:** Chair of Operations, FinTech@UW; coordinate programming, logistics, and member communication.

WORK EXPERIENCE

Morgridge Institute for Research – Stefely Lab

Sep. 2026 – Present

Student Researcher – Computational Biology

Madison, WI

- Scope Python data-processing and analysis workflows with researchers to study mitochondrial protein complexes using mass spectrometry data.

United Nations Development Programme (UNDP)

Dec. 2025 – Jan. 2026

Independent Research Consultant

Doha, Qatar

- Built a reusable Python/pandas pipeline to clean and standardize OECD aid-flow records, enabling comparisons of donor allocations across Qatar and GCC peers.
- Created Matplotlib charts and dashboards to summarize allocation patterns for internal strategy discussions.

MD Anderson Cancer Center

Jun. 2025 – Aug. 2025

Bioinformatics Engineering Intern

Houston, TX

- Reduced Linux HPC setup from approximately **2.5 hours** to under **15 minutes** by automating installation with Bash and documenting a repeatable setup procedure.
- Containerized NetMHCstabpan with Docker, packaging legacy Linux dependencies for reproducible deployment and documenting AMD64 emulation on Apple Silicon.

PROJECTS

Cadence | TypeScript, Next.js, FastAPI, GCP

Mar. 2026

- Co-developed a speech-fluency practice app for people with speech impediments, combining Whisper transcription, Wav2Vec2 embeddings, and classification to identify disfluencies in recordings.
- Deployed a Dockerized FastAPI backend to Google Cloud Run and connected a TypeScript/Next.js interface with Firestore session storage and audio-timeline feedback.

Traction | PyTorch, ONNX Runtime, Streamlit

Feb. 2026

- Placed **Top 5 at the Qualcomm Edge AI Hackathon** with a team-built crop-disease ML app, deploying a PyTorch-trained MobileNetV2 model through ONNX Runtime for local camera inference.
- Incorporated Wisconsin farmer feedback into the Streamlit workflow; added detection logs and a local Ollama assistant to analyze recorded crop-disease observations.

Quant Backtest Engine | Python, pandas, NumPy, pytest

Feb. 2026

- Built a Python backtesting engine to compare **7 portfolio strategies across 12 assets**, with Parquet data caching, transaction-cost modeling, and reusable strategy interfaces.
- Added walk-forward evaluation and pytest coverage for portfolio accounting, missing-data handling, and look-ahead bias to check that historical decisions exclude future data.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C, C++, TypeScript, JavaScript, Bash

Web / Databases: React, Next.js, FastAPI, PostgreSQL, Firestore

ML / Data: PyTorch, scikit-learn, ONNX Runtime, pandas, NumPy, DuckDB, Parquet

Tools: Git, Linux, Docker, Google Cloud, pytest, Ollama, Claude Code